

Minimal Lawful Move Law for HSF Mesoscopic Churn

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Abstract

This note defines the minimal lawful move law for the next HSF mesoscopic implementation. Its purpose is to prevent further drift back into hand-coded birth/prune engines. The note freezes the admissible move set, the local comparison region, the provisional move score, and the derivation status of each term. The intended use is simple: this note should be the specification for the next script, and the script should do no more than what this note allows.

1 Purpose

The current HSF mesoscopic work has already established several things:

1. A churn rule must be based on one local move principle, not separate hand-written event laws.
2. No-refolding must be treated as protection of committed collective relief structure rather than generic graph persistence.
3. The next implementation must separate *derived structure*, *temporary proxy*, and *open term*.

This note is therefore not another interpretive essay. It is a decision note.

2 Frozen admissible move set

For the next lawful implementation, the admissible local move set is frozen as

$$\mathcal{M}_R = \{M_0, M_+, M_{e-}, M_-\}, \tag{1}$$

where:

M_0 : no move,
 M_+ : birth one explicit support patch,
 M_{e-} : weaken one committed interface / link,
 M_- : demote one explicit support patch.

No other move classes are to be added in the next implementation.

Decision status

- Derived enough to freeze now.

3 Frozen local comparison region

For any candidate move, let R be the local comparison region defined by:

1. the candidate parent interface, candidate link, or candidate support patch,
2. its immediate graph-neighborhood,
3. and any overlapping committed shell/core structure if present.

Equivalently, R is the minimal mesoscale neighborhood needed to ask whether the move preserves or improves the local collective relief function.

Decision status

- Derived enough to freeze now.

4 Counterfactual move evaluation

Each candidate move $M \in \mathcal{M}_R$ is evaluated by branch comparison.

4.1 Baseline branch

Construct a baseline branch in which no change is made to the current local support/interface arrangement in R , and evolve it for a short fixed horizon τ .

4.2 Moved branch

Construct a moved branch in which M is applied locally in R , and evolve it for the same short fixed horizon τ .

4.3 Move score

The move score is

$$\Delta\mathcal{F}_R(M) = \mathcal{F}_R^{\text{moved}}(\tau) - \mathcal{F}_R^{\text{base}}(\tau). \quad (2)$$

The selected move is

$$M^* = \arg \max_{M \in \mathcal{M}_R} \Delta\mathcal{F}_R(M), \quad (3)$$

and M^* is accepted only if

$$\Delta\mathcal{F}_R(M^*) > 0. \quad (4)$$

Decision status

- Derived enough to freeze now.

5 Minimal lawful move functional

The minimal lawful local move functional is

$$\boxed{\Delta\mathcal{F}_R(M) = \Delta\mathcal{E}_{\text{Rexpr}}(M) - \lambda_B \Delta\mathcal{C}_R^{(B)}(M) - \lambda_S \Delta\mathcal{C}_R^{(S)}(M) - \lambda_F \Delta\mathcal{C}_R^{(F)}(M) - \lambda_R W_{\text{NR}}(M)} \quad (5)$$

with all $\lambda_* \geq 0$.

This is the *minimal lawful form*. The next implementation should not add new term classes beyond these.

6 Meaning of the term classes

6.1 Expression gain $\Delta\mathcal{E}_{\text{Rexpr}}(M)$

This term rewards a move when it improves the local constrained expression of the wavefunction.

For the next implementation, the minimal expression gain is

$$\Delta\mathcal{E}_{\text{Rexpr}}(M) = w_{\text{MI}} \Delta E_{\text{MI}}(M) + w_{\text{corr}} \Delta E_{\text{corr}}(M) + w_{\text{link}} \Delta E_{\text{link}}(M), \quad (6)$$

where:

- $\Delta E_{\text{MI}}(M)$: change in local retained mutual-information structure,
- $\Delta E_{\text{corr}}(M)$: change in parent/core correlator strength,
- $\Delta E_{\text{link}}(M)$: change in link/interface activity quality.

Derivation status

- ΔE_{MI} : **temporary proxy, lawful to use now**
- ΔE_{corr} : **temporary proxy, lawful to use now**
- ΔE_{link} : **temporary proxy, lawful to use now**
- exact linear combination: **open**

6.2 Finite-bandwidth burden $\Delta\mathcal{C}_R^{(B)}(M)$

This term penalizes moves that increase active-channel burden, overstuffed interfaces, or unnecessary explicit-support load.

For the next implementation, the preferred proxy family is influence-map based:

$$\Delta\mathcal{C}_R^{(B)}(M) \sim \Delta\left(\text{effective active-channel rank} / \text{effective rank} / \text{active-channel occupancy}\right). \quad (7)$$

Fallback occupancy terms such as raw support count or raw edge count may be used only as subordinate burden surrogates, not as the primary meaning of bandwidth.

Derivation status

- influence-map rank/effective-rank family: **temporary proxy, lawful to use now**
- raw node/edge counts as primary bandwidth meaning: **not lawful as primary term**
- exact bandwidth functional: **open**

6.3 Spread / no-signaling burden $\Delta\mathcal{C}_R^{(S)}(M)$

This term penalizes moves that make support/influence spread unnecessarily or too broadly.

For the next implementation, the preferred proxy family is also influence-map based:

$$\Delta\mathcal{C}_R^{(S)}(M) \sim \Delta\left(\text{channel spread} / \text{effective-rank spread} / \text{local influence breadth}\right). \quad (8)$$

Graph-embedding spread may be used only as a secondary diagnostic, not as the primary lawful term.

Derivation status

- influence-map spread family: **temporary proxy, lawful to use now**
- spectral 1D extent as primary term: **not lawful as primary term**
- exact spread functional: **open**

6.4 No-forgetting penalty $\Delta\mathcal{C}_R^{(F)}(M)$

This term penalizes moves that destroy relational record rather than preserving or lawfully re-expressing it.

For the next implementation, use the minimal combination:

$$\Delta\mathcal{C}_R^{(F)}(M) = q_F (1 - F_{\text{org}}(M)) + q_I [\Delta I_{\text{lost}}(M)]_+, \quad (9)$$

where:

- $F_{\text{org}}(M) \in [0, 1]$ is organizer fidelity under the move,
- $[\Delta I_{\text{lost}}(M)]_+$ is positive local MI loss.

Derivation status

- organizer fidelity: **temporary proxy, lawful to use now**
- local MI loss: **temporary proxy, lawful to use now**
- exact recoverability-based forgetting functional: **open**

6.5 No-refolding witness $W_{\text{NR}}(M)$

This is the main protection term for no-refolding proper.

For the next implementation, the lawful no-refolding witness is the layered general witness

$$W_{\text{NR}}(M) = \alpha_{\text{ep}} W_{\text{ep}}(M) + \alpha_{\text{slack}} W_{\text{slack}}(M) + \alpha_{\text{func}} W_{\text{func}}(M), \quad (10)$$

where:

- $W_{\text{ep}}(M)$: endpoint/interface bookkeeping loss,
- $W_{\text{slack}}(M)$: slack creation,
- $W_{\text{func}}(M)$: relief-function loss.

The next implementation should not replace this by generic shell loss, triangle loss, edge persistence, or graph-change penalties.

Derivation status

- layered witness structure: **derived enough to freeze now**
- exact endpoint map and overlap: **open**
- exact slack functional: **temporary proxy, lawful to use now**
- exact function-loss decomposition: **temporary proxy, lawful to use now**

7 Sector-specific add-ons

The general move law must remain general. Sector-specific bookkeeping terms are added only when the sector is physically relevant.

7.1 Fermionic sector

If fermionic/Jordan–Wigner bookkeeping is explicitly represented, use

$$\Delta\mathcal{F}_{R\text{ferm}}(M) = \Delta\mathcal{F}_R(M) - \lambda_{\text{JW}} W_{\text{JW}}(M). \quad (11)$$

Derivation status

- sector-specific add-on concept: **derived enough to freeze now**
- exact W_{JW} : **open**

7.2 Gauge/interface sector

If local compatibility or Gauss-like attachment bookkeeping is explicitly represented, use

$$\Delta\mathcal{F}_{R\text{gauge}}(M) = \Delta\mathcal{F}_R(M) - \lambda_{\Gamma} W_{\text{gauge}}(M). \quad (12)$$

Derivation status

- sector-specific add-on concept: **derived enough to freeze now**
- exact W_{gauge} : **open**

7.3 Mesoscopic object sector

If collective-mode identity is explicitly being preserved, use

$$\Delta\mathcal{F}_{R\text{obj}}(M) = \Delta\mathcal{F}_R(M) - \lambda_{\text{mode}} W_{\text{mode}}(M). \quad (13)$$

Derivation status

- sector-specific add-on concept: **derived enough to freeze now**
- exact W_{mode} : **open**

8 Minimal lawful implementation policy

The next script is lawful only if it obeys all of the following:

1. It uses only the frozen move set $\{M_0, M_+, M_{e-}, M_-\}$.
2. It uses local counterfactual branch comparison.
3. It uses the minimal move law (5) or a strict sector-specific extension of it.
4. It labels all nonderived observables as temporary proxies in code comments and notes.
5. It does not silently replace W_{NR} with graph-persistence heuristics.
6. It does not present any chosen weights as derived constants.

9 What is forbidden in the next implementation

The following are forbidden as primary move-law terms:

- shell size loss as the meaning of no-refolding,
- raw edge persistence as the meaning of commitment,
- spectral extent as the primary no-signaling burden,
- basis-state return as the primary no-forgetting term,
- any new move class beyond the frozen move set.

These may still be logged as diagnostics, but not promoted into the core move law.

10 Status table

Move-law component	Status	Implementation note
Frozen move set	Derived enough now	Must be used as written.
Local comparison region R	Derived enough now	Must be local, not global.
Counterfactual branch comparison	Derived enough now	Mandatory.
Expression gain term class	Required	Use lawful temporary proxies only.
Bandwidth burden term class	Required	Prefer influence-map rank family.
Spread burden term class	Required	Prefer influence-map spread family.
No-forgetting penalty term class	Required	Use fidelity + MI-loss temporary proxies.
Layered no-refolding witness	Required	Must remain layered; do not collapse to graph persistence.
Sector-specific add-ons	Plausible / lawful when relevant	Add only if the sector is explicitly represented.
Exact coefficients / weights	Open	Must not be presented as derived.
Exact endpoint map, JW map, gauge descriptor, mode panel	Open	Leave clearly labeled as open.

11 Working conclusion

The next lawful HSF mesoscopic implementation should use one and only one minimal move law:

$$\Delta\mathcal{F}_R(M) = \Delta\mathcal{E}_{\text{Rexpr}}(M) - \lambda_B\Delta\mathcal{C}_R^{(B)}(M) - \lambda_S\Delta\mathcal{C}_R^{(S)}(M) - \lambda_F\Delta\mathcal{C}_R^{(F)}(M) - \lambda_RW_{\text{NR}}(M) \quad (14)$$

together with:

- the frozen move set,
- the frozen local comparison region,
- counterfactual branch evaluation,
- and explicit labeling of temporary proxies.

This is not yet the final derived churn law. But it is the first minimal move law that is strict enough to keep the next script honest.

One-sentence bottom line

The next lawful script should implement only a local counterfactual move law built from expression gain, bandwidth burden, spread burden, forgetting penalty, and the layered no-refolding witness—and nothing else.